

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Display settings
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', 100)

print("Libraries imported successfully!")
```

Libraries imported successfully!

```
[2]: df = pd.read_csv("traffic.csv")
```



```
print("Dataset loaded successfully!")
```

Dataset loaded successfully!

```
[3]: df.head()
```

```
[3]:
```

	event	date	country	city	artist	album	track	isrc	linkid
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
1	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
2	click	2021-08-21	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871	23199824-9cf5-4b98-942a-34965c3b0cc2
3	click	2021-08-21	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904	35573248-4e49-47c7-af80-08a960fa74cd
4	click	2021-08-21	Maldives	Malé	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8

```
4]: print("Number of rows:", df.shape[0])  
    print("Number of columns:", df.shape[1])
```

Number of rows: 226278

Number of columns: 9

```
5]: df.shape
```

```
5]: (226278, 9)
```

```
[6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   event       226278 non-null  object
1   date        226278 non-null  object
2   country     226267 non-null  object
3   city        226267 non-null  object
4   artist      226241 non-null  object
5   album       226273 non-null  object
6   track       226273 non-null  object
7   isrc        219157 non-null  object
8   linkid      226278 non-null  object
dtypes: object(9)
memory usage: 15.5+ MB
```

```
: df['date'] = pd.to_datetime(df['date'], errors='coerce')
```

```
: df['date'].dtype
```

```
: dtype('<M8[ns]')
```

```
df.describe(include='all')
```

	event	date	country	city	artist	album	track	isrc	linkid
count	226278	226278	226267	226267	226241	226273	226273	219157	226278
unique	3	NaN	211	11993	2419	3254	3562	709	3839
top	pageview	NaN	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
freq	142015	NaN	47334	22791	40841	40841	40841	40841	40841
mean	NaN	2021-08-21 21:04:31.768355584	NaN	NaN	NaN	NaN	NaN	NaN	NaN
min	NaN	2021-08-19 00:00:00	NaN	NaN	NaN	NaN	NaN	NaN	NaN
25%	NaN	2021-08-20 00:00:00	NaN	NaN	NaN	NaN	NaN	NaN	NaN
50%	NaN	2021-08-22 00:00:00	NaN	NaN	NaN	NaN	NaN	NaN	NaN
75%	NaN	2021-08-24 00:00:00	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
df.describe(include='object')
```

	event	country	city	artist	album	track	isrc	linkid
count	226278	226267	226267	226241	226273	226273	219157	226278
unique	3	211	11993	2419	3254	3562	709	3839
top	pageview	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
freq	142015	47334	22791	40841	40841	40841	40841	40841

```
] missing_values = df.isnull().sum()  
  
print(missing_values)
```

```
event      0  
date       0  
country   11  
city      11  
artist    37  
album      5  
track      5  
isrc      7121  
linkid     0  
dtype: int64
```

```
missing_percentage = (df.isnull().sum() / len(df)) * 100
```

```
missing_summary = pd.DataFrame({  
    'Missing Values': df.isnull().sum(),  
    'Missing Percentage': missing_percentage  
})
```

```
missing_summary
```

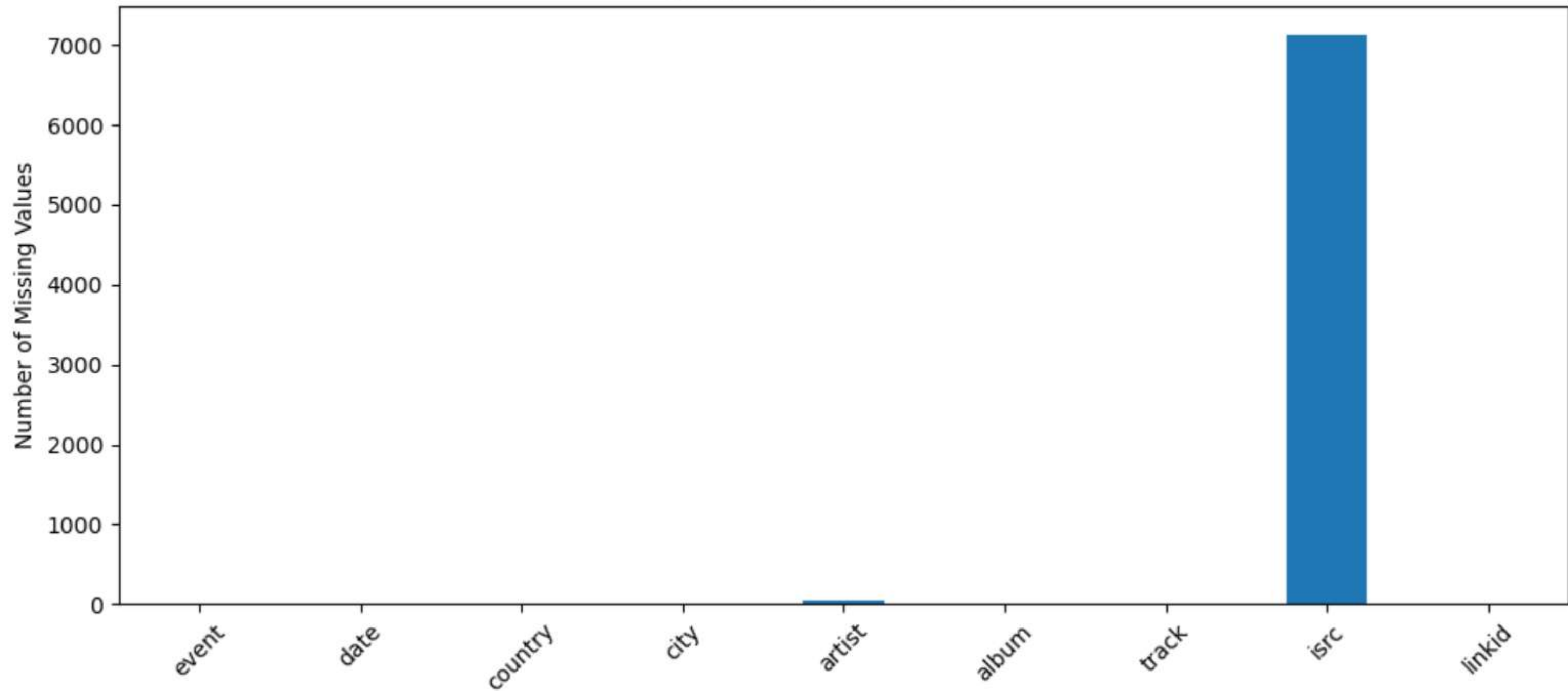
	Missing Values	Missing Percentage
event	0	0.000000
date	0	0.000000
country	11	0.004861
city	11	0.004861
artist	37	0.016352
album	5	0.002210
track	5	0.002210
isrc	7121	3.147014
linkid	0	0.000000

```
plt.figure(figsize=(10, 5))

missing_summary['Missing Values'].plot(kind='bar')

plt.title('Missing Values by Column')
plt.xlabel('Columns')
plt.ylabel('Number of Missing Values')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Missing Values by Column



```
duplicate_count = df.duplicated().sum()
```

```
print("Number of duplicate rows:", duplicate_count)
```

Number of duplicate rows: 103711

```
duplicate_percentage = (duplicate_count / len(df)) * 100
```

```
print("Duplicate percentage:", round(duplicate_percentage, 2), "%")
```

Duplicate percentage: 45.83 %

```
df = df.drop_duplicates().reset_index(drop=True)
```

```
print("Dataset shape after removing duplicates:", df.shape)
```

Dataset shape after removing duplicates: (122567, 9)

```
unique_values = df.nunique()
```

```
print(unique_values)
```

```
event          3  
date           7  
country       211  
city         11993  
artist        2419  
album         3254  
track         3562  
isrc          709  
linkid        3839  
dtype: int64
```

```
[20]: summary = pd.DataFrame({
    'Column': df.columns,
    'Data Type': df.dtypes.astype(str).values,
    'Unique Values': df.nunique().values,
    'Missing Values': df.isnull().sum().values
})
```

summary

[20]:

	Column	Data Type	Unique Values	Missing Values
0	event	object	3	0
1	date	datetime64[ns]	7	0
2	country	object	211	5
3	city	object	11993	5
4	artist	object	2419	28
5	album	object	3254	4
6	track	object	3562	4
7	isrc	object	709	6306
8	linkid	object	3839	0

```
event_counts = df['event'].value_counts()  
  
print(event_counts)
```

```
event  
pageview    73360  
click       32499  
preview     16708  
Name: count, dtype: int64
```

```
plt.figure(figsize=(8, 5))
```

```
event_counts.plot(kind='bar')
```

```
plt.title('Distribution of Website Events')
```

```
plt.xlabel('Event Type')
```

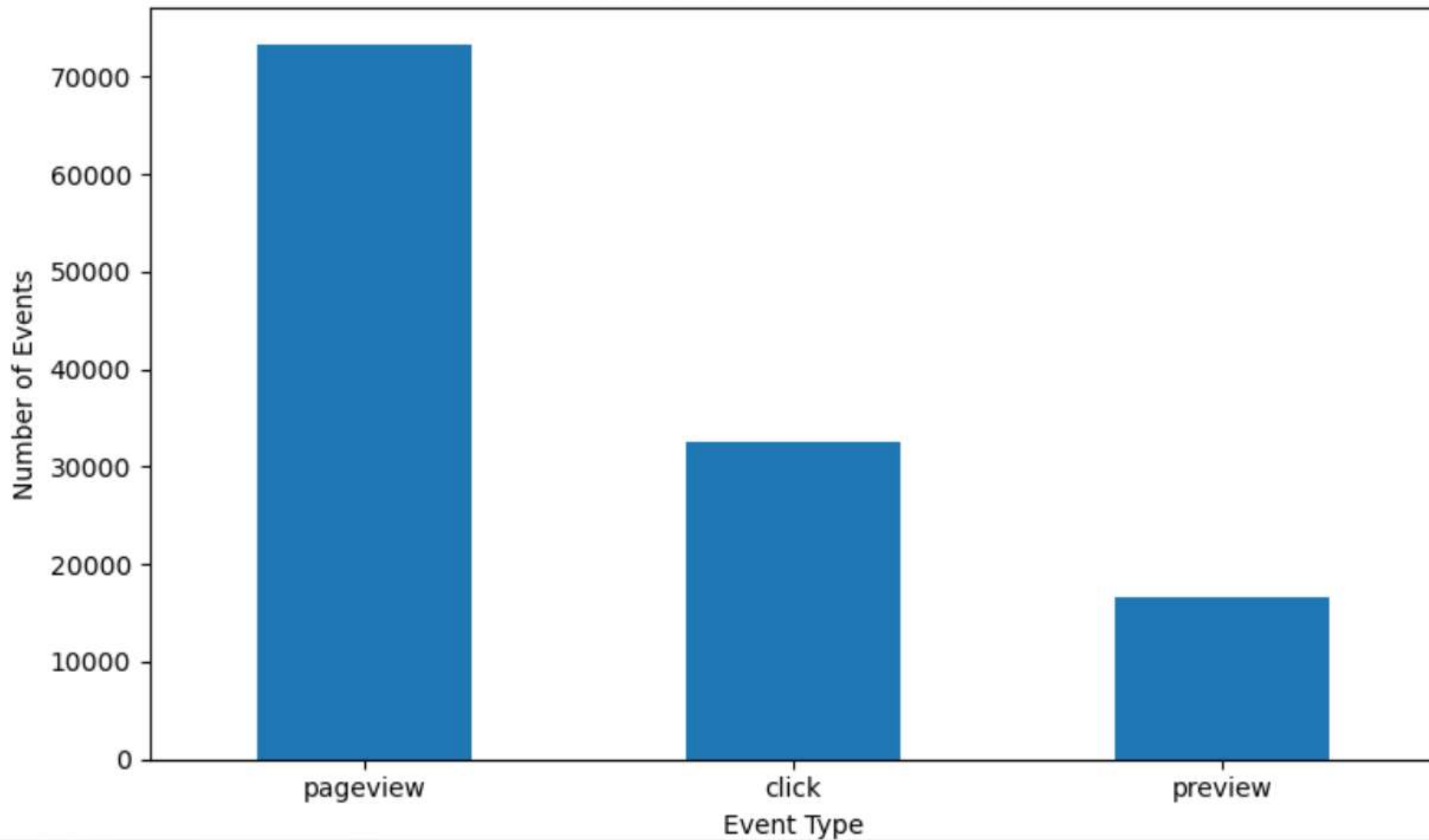
```
plt.ylabel('Number of Events')
```

```
plt.xticks(rotation=0)
```

```
plt.tight_layout()
```

```
plt.show()
```

Distribution of Website Events



```
: print("Minimum date:", df['date'].min())  
print("Maximum date:", df['date'].max())
```

Minimum date: 2021-08-19 00:00:00

Maximum date: 2021-08-25 00:00:00

```
: daily_events = df.groupby('date').size()
```

`daily_events`

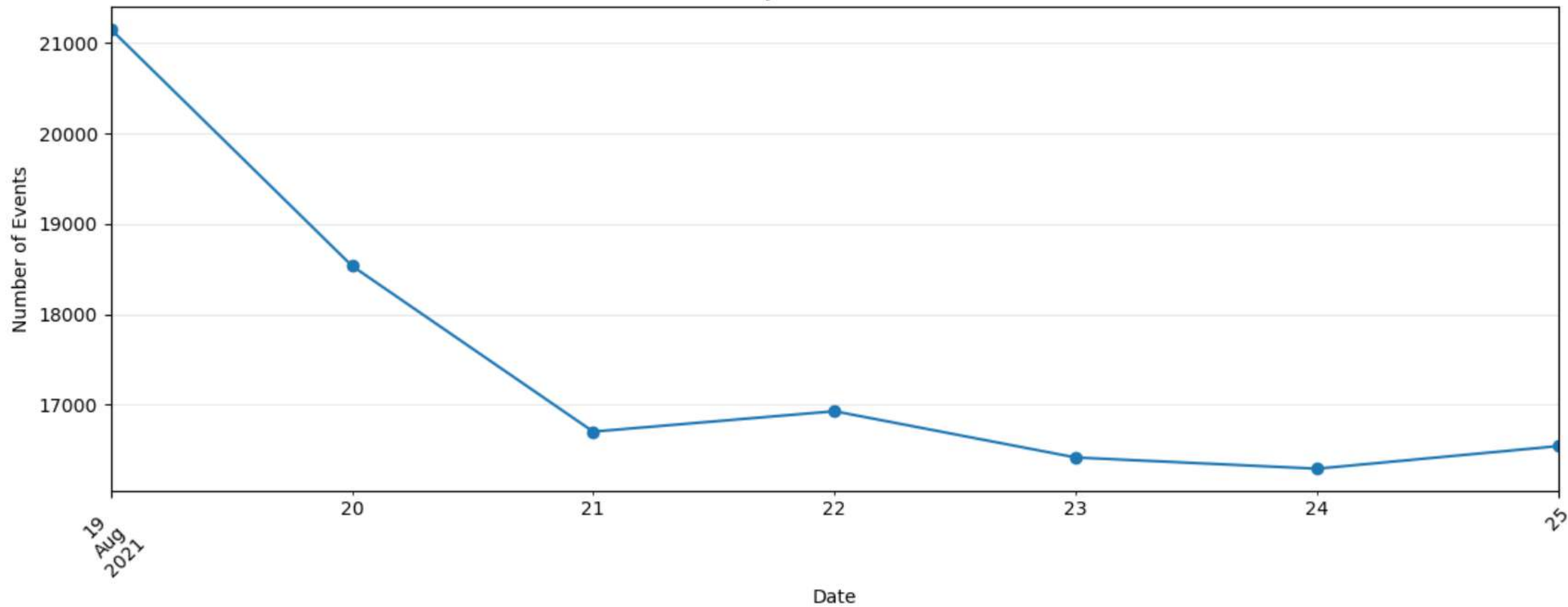
```
: date  
2021-08-19    21156  
2021-08-20    18536  
2021-08-21    16701  
2021-08-22    16927  
2021-08-23    16415  
2021-08-24    16291  
2021-08-25    16541  
dtype: int64
```

```
plt.figure(figsize=(12, 5))

daily_events.plot(kind='line', marker='o')

plt.title('Daily Website Traffic')
plt.xlabel('Date')
plt.ylabel('Number of Events')
plt.xticks(rotation=45)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

Daily Website Traffic



```
daily_event = df.groupby(['date', 'event']).size().unstack(fill_value=0)
```

daily_event

event	click	pageview	preview
-------	-------	----------	---------

date

2021-08-19	5860	13006	2290
------------	------	-------	------

2021-08-20	4999	11088	2449
------------	------	-------	------

2021-08-21	4279	9939	2483
------------	------	------	------

2021-08-22	4421	9982	2524
------------	------	------	------

2021-08-23	4314	9722	2379
------------	------	------	------

2021-08-24	4231	9705	2355
------------	------	------	------

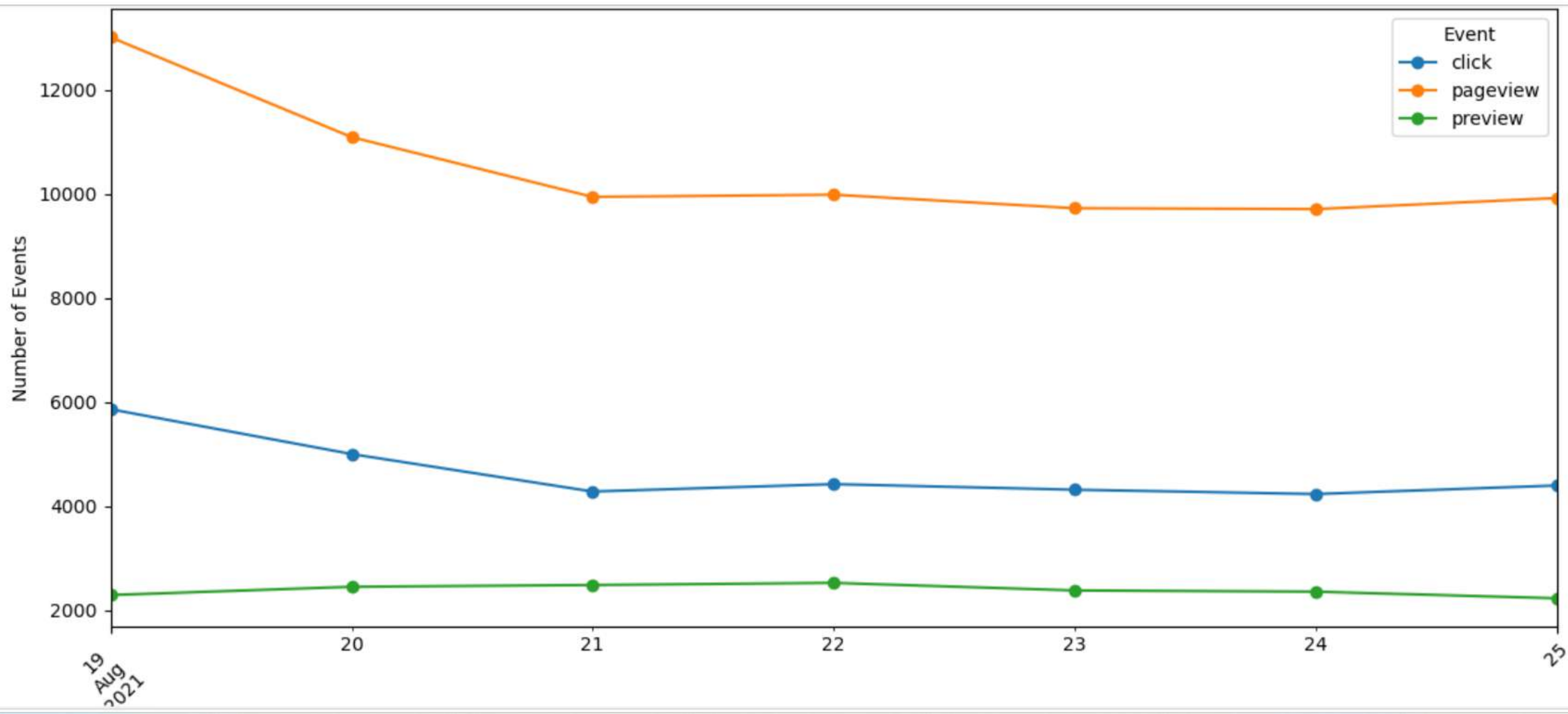
2021-08-25	4395	9918	2228
------------	------	------	------

```
plt.figure(figsize=(12, 6))

daily_event.plot(kind='line', marker='o', figsize=(12, 6))

plt.title('Daily Traffic by Event Type')
plt.xlabel('Date')
plt.ylabel('Number of Events')
plt.xticks(rotation=45)
plt.legend(title='Event')
plt.tight_layout()
plt.show()
```

<Figure size 1200x600 with 0 Axes>



```
top_countries = df['country'].value_counts().head(10)
```

```
top_countries
```

```
country
```

```
United States    28664
```

```
India            18689
```

```
France           10565
```

```
Saudi Arabia     7682
```

```
United Kingdom   5095
```

```
Germany          4017
```

```
Canada           2784
```

```
Pakistan         2633
```

```
Iraq             2444
```

```
Turkey           2399
```

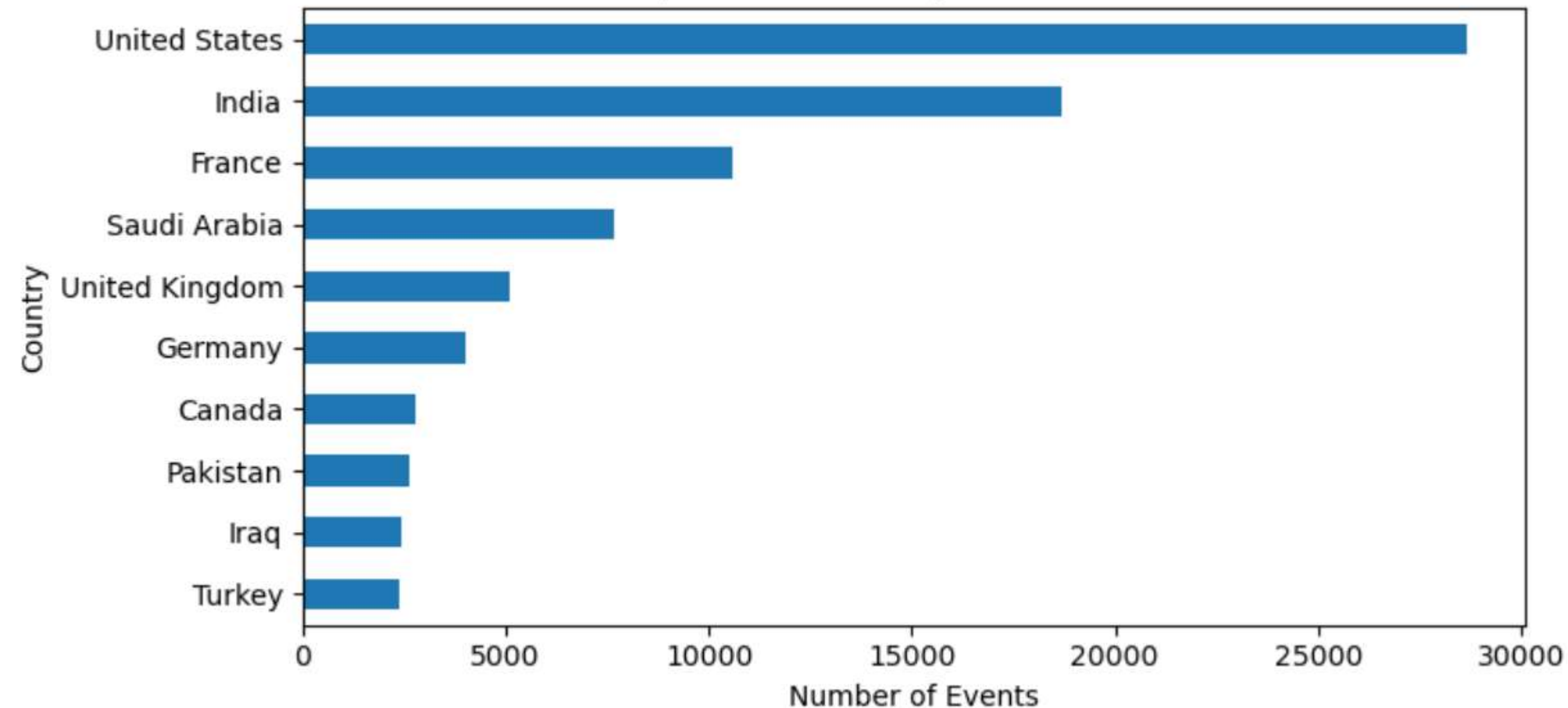
```
Name: count, dtype: int64
```

```
plt.figure(figsize=(8, 4))

top_countries.sort_values().plot(kind='barh')

plt.title('Top 10 Countries by Website Traffic')
plt.xlabel('Number of Events')
plt.ylabel('Country')
plt.tight_layout()
plt.show()
```

Top 10 Countries by Website Traffic



```
top_cities = df['city'].value_counts().head(10)
```

```
top_cities
```

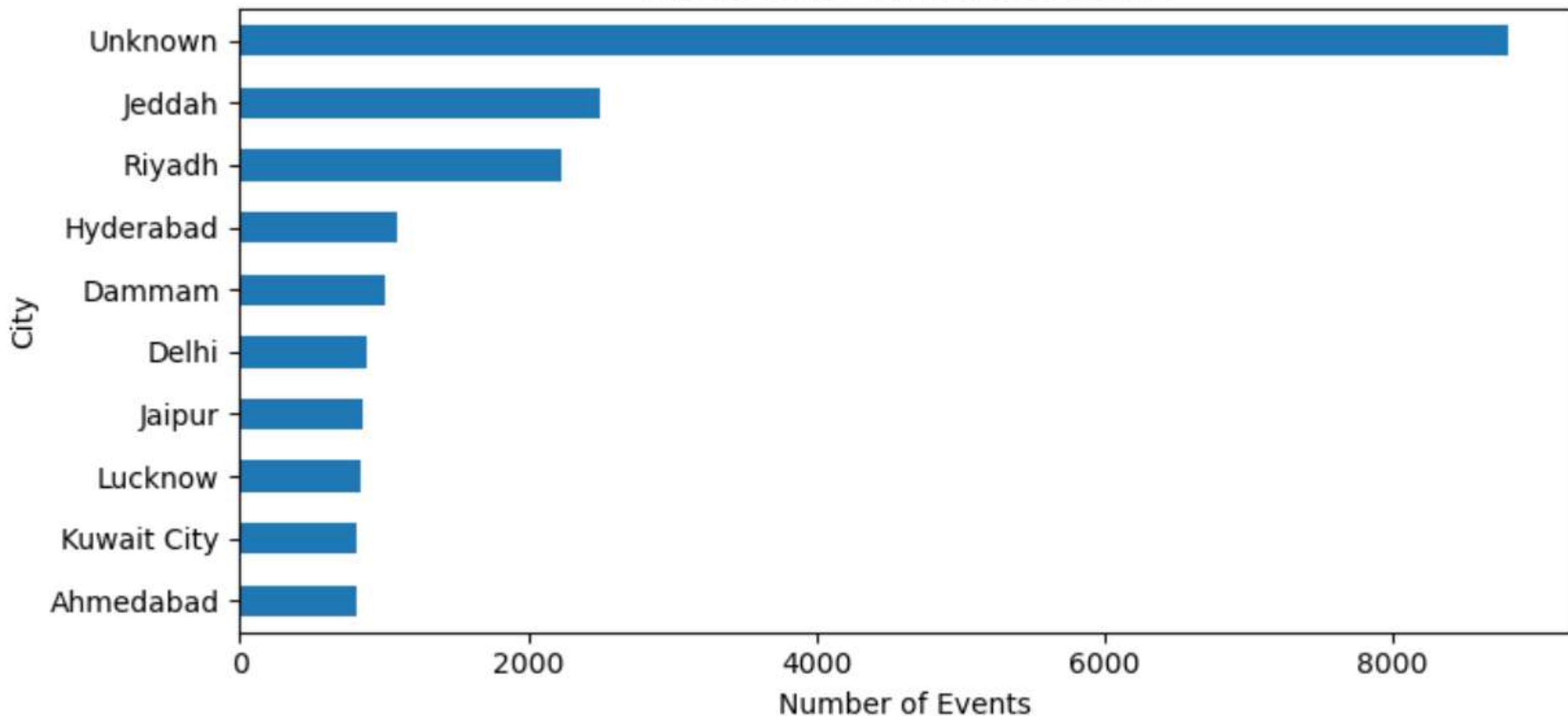
```
city
Unknown      8797
Jeddah       2497
Riyadh       2232
Hyderabad    1088
Dammam       1002
Delhi        884
Jaipur       849
Lucknow      837
Kuwait City  816
Ahmedabad    808
Name: count, dtype: int64
```

```
: plt.figure(figsize=(8, 4))

top_cities.sort_values().plot(kind='barh')

plt.title('Top 10 Cities by Website Traffic')
plt.xlabel('Number of Events')
plt.ylabel('City')
plt.tight_layout()
plt.show()
```

Top 10 Cities by Website Traffic



```
top_artists = df['artist'].value_counts().head(10)
```

```
top_artists|
```

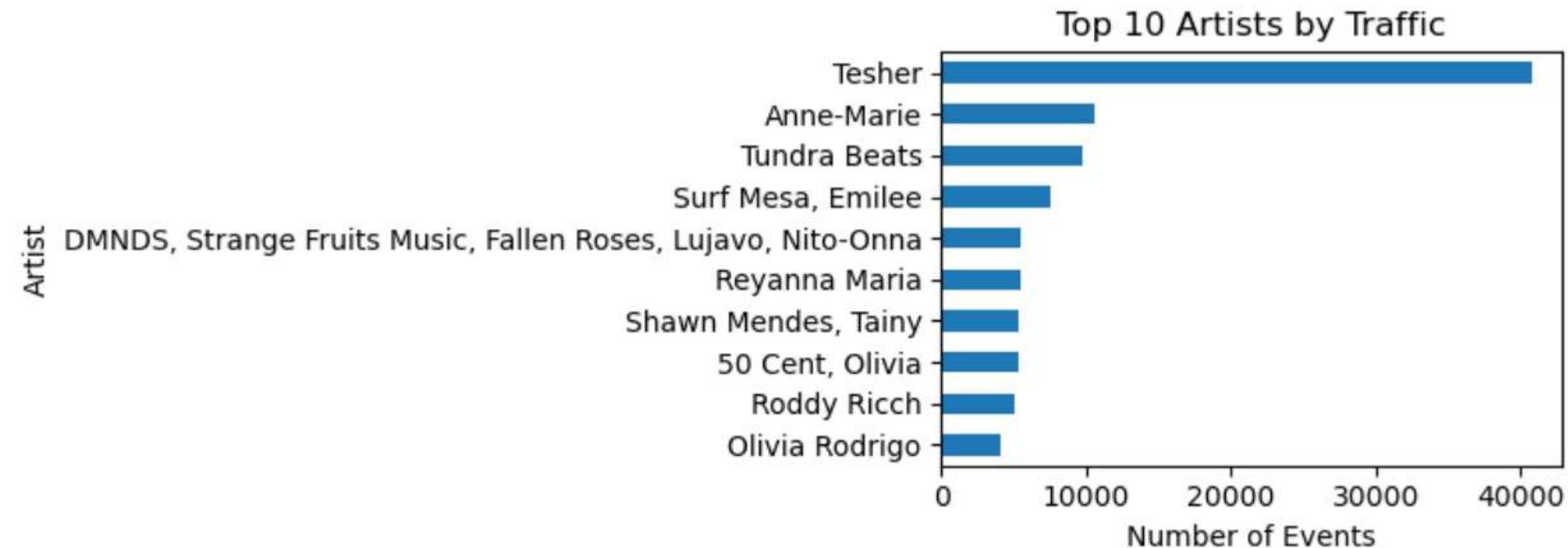
artist	
Tesher	40841
Anne-Marie	10650
Tundra Beats	9751
Surf Mesa, Emilee	7533
DMNDS, Strange Fruits Music, Fallen Roses, Lujavo, Nito-Onna	5512
Reyanna Maria	5437
Shawn Mendes, Tainy	5409
50 Cent, Olivia	5367
Roddy Ricch	5093
Olivia Rodrigo	4115

Name: count, dtype: int64

```
plt.figure(figsize=(8, 3))

top_artists.sort_values().plot(kind='barh')

plt.title('Top 10 Artists by Traffic')
plt.xlabel('Number of Events')
plt.ylabel('Artist')
plt.tight_layout()
plt.show()
```



```
top_tracks = df['track'].value_counts().head(10)
```

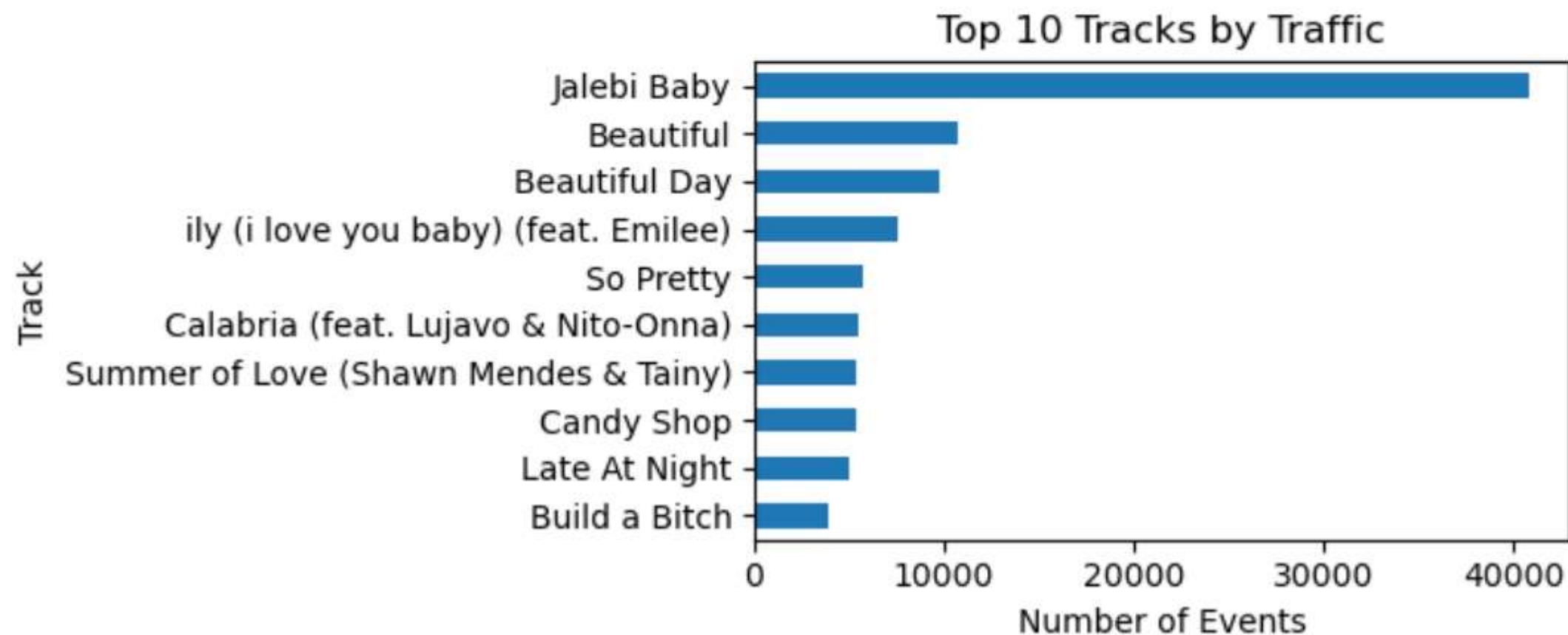
```
top_tracks
```

```
track
Jalebi Baby                40841
Beautiful                  10658
Beautiful Day               9751
ily (i love you baby) (feat. Emilee)  7533
So Pretty                  5664
Calabria (feat. Lujavo & Nito-Onna)  5512
Summer of Love (Shawn Mendes & Tainy)  5409
Candy Shop                 5375
Late At Night              5036
Build a Bitch              3879
Name: count, dtype: int64
```

```
plt.figure(figsize=(7, 3))

top_tracks.sort_values().plot(kind='barh')

plt.title('Top 10 Tracks by Traffic')
plt.xlabel('Number of Events')
plt.ylabel('Track')
plt.tight_layout()
plt.show()
```



```
: unique_links = df['linkid'].nunique()

print("Unique links:", unique_links)
```

Unique links: 3839

```
: top_links = df['linkid'].value_counts().head(10)
```

top_links

```
: linkid
2d896d31-97b6-4869-967b-1c5fb9cd4bb8    40841
522da5cc-8177-4140-97a7-a84fdb4caf1c    10314
e849515b-929d-44c8-a505-e7622f1827e9     9750
c2c876ab-b093-4750-9449-6b4913da6af3     6733
681d888a-59ce-4acb-b7c5-95dab0c063d9     5512
23199824-9cf5-4b98-942a-34965c3b0cc2     5437
97105aff-3e40-4c2e-bfeb-563b55e8f5dd     5367
d5317591-872a-487d-87cb-10a516643d5f     5352
a82e72d8-523f-43ab-9fd0-992605d54b4e     5036
a107c329-5d35-4c7d-8050-50bc69ce7dbc     3879
Name: count, dtype: int64
```

```
categorical_columns = [  
    'event',  
    'country',  
    'city',  
    'artist',  
    'album',  
    'track',  
    'isrc',  
    'linkid'  
]  
  
for col in categorical_columns:  
    blank_count = (df[col].astype(str).str.strip() == '').sum()  
    print(f"{col}: {blank_count} blank values")
```

```
event: 0 blank values  
country: 0 blank values  
city: 0 blank values  
artist: 0 blank values  
album: 0 blank values  
track: 0 blank values  
isrc: 0 blank values  
linkid: 0 blank values
```

```
print("Final Dataset Shape:")
print(df.shape)

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:")
print(df.duplicated().sum())
```

Final Dataset Shape:
(226278, 9)

Data Types:

event	object
date	object
country	object
city	object
artist	object
album	object
track	object
isrc	object
linkid	object
dtype:	object

Missing Values:

event	0
date	0
country	11
city	11
artist	37
album	5
track	5
isrc	7121
linkid	0
dtype:	int64

Duplicate Rows:

103711

```
df.to_csv("traffic_cleaned.csv", index=False)
```

```
print("Cleaned dataset saved successfully!")
```

```
Cleaned dataset saved successfully!
```